

1. Stepper Motor Positioning Principle

The robotic arm joints are actuated using **NEMA 17 stepper motors (42BYGH60-01A)**. Stepper motors rotate in fixed angular increments (typically 1.8° per step for a 200-step motor). Therefore, joint position can be determined by counting the number of step pulses sent to the motor.

For a 200-step motor:

$$\text{Step Angle} = \frac{360^\circ}{200} = 1.8^\circ$$

Joint angular displacement is calculated as:

$$\theta = \text{Step Count} \times 1.8^\circ$$

This method allows open-loop position tracking based purely on commanded steps. The stepper motor is not directly connected to the robotic joint. Instead, motion is transmitted through a system of **gears and/or belt drives**. Therefore, the joint angular displacement is not equal to the motor angular displacement and must be determined using the transmission ratio.

2. Homing and Reference Position

At system startup, the controller does not know the physical position of the motor shaft. Therefore, a **limit switch** is installed at each joint to define a mechanical home position.

Homing procedure:

1. The motor moves slowly toward the home direction.
2. When the limit switch is triggered, motion stops.
3. The controller sets the position counter to zero.
4. All subsequent motion is referenced to this known position.

This ensures consistent and repeatable positioning after every power cycle.

3. Encoder as an Alternative or Backup Option

Although step counting provides a simple position tracking method, it is based on the assumption that no steps are missed.

Potential issues with open-loop stepper control include:

- Step loss under high load
- Mechanical vibration
- Unexpected external disturbances

To prevent such issues, a **rotary encoder** can be added to provide physical position feedback.

Possible implementation options include:

- Using a **single rotary encoder** mounted on the motor shaft to verify actual position.
- Comparing the encoder reading with the expected step count to detect missed steps.
- Using **two encoders** (for example, one on the motor shaft and one on the joint output shaft) for increased reliability and backlash monitoring.
- Comparing encoder feedback with the internally tracked motor position for error correction.

The encoder-based solution would convert the system from open-loop to closed-loop control if required.

At this stage, the encoder is considered an optional enhancement depending on observed system performance during testing.

4. Motion Control Software (Under Evaluation)

The use of the **AccelStepper library** is currently being evaluated to simplify stepper motor control and improve motion smoothness.

A stepper motor moves in discrete steps, and its position can be tracked by counting the number of step pulses sent from the controller. This position tracking can be implemented manually in software without the use of external libraries. In its simplest form, the program increments or decrements a position variable every time a step pulse is generated.

However, generating precise step pulses manually requires careful timing control. If step signals are issued too quickly or without proper acceleration control, the motor may:

- Miss steps
- Vibrate excessively
- Produce jerky motion
- Lose synchronization with the internal step counter

The AccelStepper library assists by automatically handling:

- Acceleration and deceleration profiles
- Smooth speed transitions
- Timing of step pulses
- Internal software-based step counting
- Non-blocking control for multiple motors

It is important to note that the library does not physically measure motor position. It only tracks position internally by counting the steps it commands. If steps are missed due to

overload or mechanical issues, the library cannot detect this error without additional feedback such as an encoder.

Therefore, while the library simplifies motion control and improves smoothness, it does not eliminate the need for homing procedures or optional encoder feedback if higher reliability is required.

Final implementation of the library will depend on performance testing and overall system stability.

<https://forum.arduino.cc/t/stepper-motor-position-control/683692>

<https://www.airspayce.com/mikem/arduino/AccelStepper/>